TMUA / PRACTICE PAPER

Set A Paper 1

Applications of Mathematical Knowledge 20 questions · 75 minutes

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Exponentials & Logarithms

1. Find the complete set of positive real values of x, with $x \neq 1$, for which

$$\log_2 x < \log_x 2$$

A $0 < x < \frac{1}{2}$ or $1 < x < 2$

B $\frac{1}{2} < x < 1$ or $x > 2$

C $1 < x < 2$

D $0 < x < \frac{1}{2}$

E $0 < x < 1$ or $x > 2$

F $x > 2$

G $0 < x < 2$, $x \neq 1$

Inequalities

2. Fix a real number $a \neq 0$.

Consider the inequality

$$|x - a| + |x + a| < 3|a|$$

Which of the following describes the set of all real x satisfying the inequality?

A No real solutions

B All real x with $|x| < |a|$

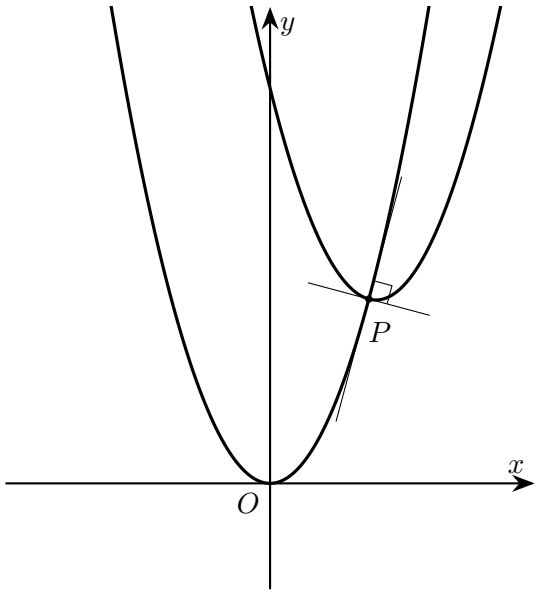
C All real x with $|x| < \dfrac{3|a|}{2}$

D All real x with $|x| < 3|a|$

E All real x with $\dfrac{|a|}{2} < |x| < \dfrac{3|a|}{2}$

Differentiation

3. The diagram shows parts of the curves $y = x^2$ and $y = (x-2)^2 + b$, where b is positive.
The curves meet at P, and the tangents to the curves at P are perpendicular.
What is the value of b?



A $\frac{\sqrt{3}}{2}$

B $\sqrt{3}$

C $2\sqrt{3}$

D $3\sqrt{3}$

E $4\sqrt{3}$

Algebra

4. A positive real number x satisfies

$$(x-2)(x+3)(x+4)(x+9) = 3024$$

What is the value of $x^2 + 7x$?

A 16

B 38

C 54

D 60

E 114

Integration

5. The curve $y = x^3 - 3x + 5$ has a tangent line at the point where $x = 2$.
This tangent meets the curve again at exactly one other point.
Find the area of the region enclosed between the curve and the tangent.

A -108

B 54

C 96

D 108

E 120

F 216

Coordinate Geometry

6. A straight line passes through the point $(2, 5)$ and has gradient m.

The line forms a triangle of area 25 with the positive x-axis and the positive y-axis.

What is the sum of all possible values of m?

A -15

B $-\frac{15}{2}$

C $-\frac{25}{4}$

D $-\frac{15}{4}$

E $\frac{15}{2}$

Number

7. Consider the integer

$$N = 8^{15} \times 5^{40}$$

When N is written out in full, how many digits does it have?

A 40

B 41

C 42

D 43

E 46

F 55

Sequences & Series

8. The nth term of a sequence is a quadratic expression in n.
The second, fifth and eighth terms are 4, 13 and 40, respectively.
What is the eleventh term?

A 67

B 76

C 85

D 94

E 121

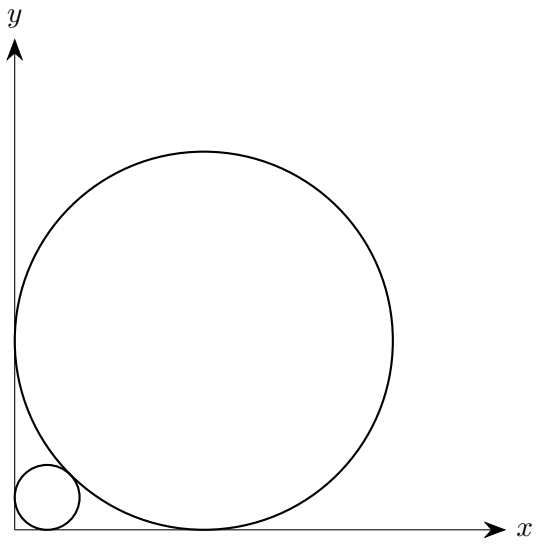
Geometry

9. A circle of radius 1 lies in the region where $x \geq 0$ and $y \geq 0$, and touches both axes.

A second, smaller circle also lies in this region, touches both axes, and touches the first circle at exactly one point.

The two circles do not overlap.

What is the radius of the smaller circle?



A $\frac{1}{3}$

B $3 - 2\sqrt{2}$

C $\sqrt{2} - 1$

D $\frac{\sqrt{2} - 1}{2}$

E $2 - \sqrt{2}$

F $17 - 12\sqrt{2}$

G $3 + 2\sqrt{2}$

Sequences & Series

10. A sequence a_n is defined by $a_0 = 3$ and, for $n \geq 0$,

$$a_{n+1} = 3(a_n)^2$$

Which of the following is equal to $\log_3(a_n)$?

A 2^n

B $2^n - 1$

C $2^{n+1} - 1$

D $2^{n+1} + 1$

E $2^n - 2$

F $2^{n+1} - 2$

Number

11. An integer n is chosen at random from the integers 1 to 1260 inclusive, with each integer equally likely to be chosen.

The fraction $\dfrac{n}{1260}$ is written as a decimal.

What is the probability that this decimal terminates and that its shortest form has exactly two digits after the decimal point?

In the shortest form, zeros at the end are omitted; for example, 0.50 is written as 0.5.

A $\dfrac{1}{630}$

B $\dfrac{2}{315}$

C $\dfrac{1}{126}$

D $\dfrac{1}{90}$

E $\dfrac{1}{70}$

F $\dfrac{1}{63}$

Functions & Graphs

12. A function f has the form $f(x) = ax + b$, where a and b are real constants.
For every real x, applying f twice gives

$$f(f(x)) = 4x + 3$$

What is the difference between the greatest and least possible values of $f(2)$?

A 2

B 4

C 7

D 10

E 12

F 14

Algebra

13. The equation $x^2 - 7x + 1 = 0$ has two positive solutions, p and q.

Find the value of

$$\frac{p-1}{p+1} \times \frac{q-1}{q+1}$$

A $-\frac{2}{3}$

B $-\frac{5}{8}$

C $-\frac{5}{9}$

D $\frac{5}{9}$

E $\frac{5}{8}$

F $\frac{2}{3}$

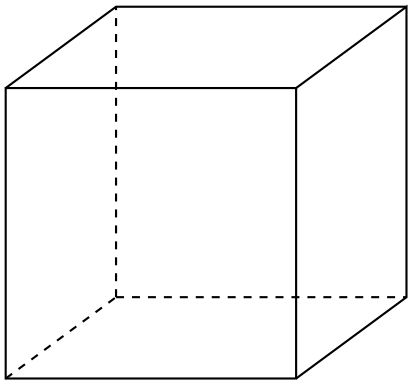
Probability & Counting

14. A cube has edge length 1.

Three different vertices are chosen at random.

Every choice of three vertices is equally likely.

What is the probability that the three chosen vertices form a right-angled triangle?



A $\frac{3}{7}$

B $\frac{1}{2}$

C $\frac{4}{7}$

D $\frac{3}{4}$

E $\frac{6}{7}$

F 1

Probability & Counting

15. A box contains red counters and blue counters. The number of red counters exceeds the number of blue counters by an integer d.

Two counters are selected at random, one after the other, without replacement. The probability that the counters have the same colour is equal to the probability that they have different colours.

In terms of d, how many counters are in the box?

A $\dfrac{d^2}{2}$

B $d(d-1)$

C d^2

D $d(d+1)$

E $2d^2$

Polynomials

16. Consider the curve C with equation

$$y = x^3 - 3x^2 + 1$$

Find the complete set of values of the constant c for which there are exactly three distinct lines passing through the point $(0, c)$ that are tangent to C.

A $-3 < c < 1$

B $-3 \leq c \leq 1$

C $1 < c < 2$

D $1 \leq c \leq 2$

E $c < -3$ or $c > 1$

F $c < 1$ or $c > 2$

Trigonometry

17. Find the complete set of real values of k for which the equation

$$2\cos^2 x - k\sin x + k = 0$$

has exactly one solution in the range $0 \leq x < 2\pi$.

A $k < -4$ or $k > 0$

B $k \leq -4$ or $k > 0$

C $k < -4$ or $k \geq 0$

D $k \leq -4$ or $k \geq 0$

E $-4 < k < 0$

F $-4 \leq k \leq 0$

Polynomials

18. Let $f(x) = x^4 + ax^2 + b$, where a and b are real constants.

The minimum value of $f(x)$ is -4, and the equation $f(x) = 0$ has exactly three distinct real solutions.

What is the value of a and the value of b?

A $a = -4$, $b = -4$

B $a = -4$, $b = 0$

C $a = -2$, $b = 0$

D $a = 0$, $b = -4$

E $a = 4$, $b = -4$

F $a = 4$, $b = 0$

Functions & Graphs

19. For a positive real constant k, the equation

$$\left|x^2 - 4x + 3\right| = kx$$

has exactly three distinct real solutions.

Find the complete set of possible values of k.

A There are no such positive values of k.

B $0 < k < 4 - 2\sqrt{3}$

C $k = 4 - 2\sqrt{3}$

D $k = 4 + 2\sqrt{3}$

E $k = 4 - 2\sqrt{3}$ or $k = 4 + 2\sqrt{3}$

Geometry

20. A circle passes through the vertices of a right-angled triangle ABC, where angle $ABC = 90°$, $AB = 12$ cm, and $BC = 16$ cm.

A line is drawn through A parallel to BC.

This line intersects the tangent to the circle at B at the point P.

What is the length of AP?

A 7.2 cm

B 8 cm

C 9 cm

D 9.6 cm

E 10 cm

F 16 cm